Invited: Physical Design for Advanced 3D ICs: Challenges and Solutions

Yuxuan Zhao
The Chinese University of Hong Kong
Hong Kong SAR
yxzhao21@cse.cuhk.edu.hk

Lancheng Zou
The Chinese University of Hong Kong
Hong Kong SAR
lczou23@cse.cuhk.edu.hk

Bei Yu
The Chinese University of Hong Kong
Hong Kong SAR
byu@cse.cuhk.edu.hk

Abstract

As technology scaling predicted by Moore's law slows down, 3D integrated circuits (3D ICs) have emerged as a promising alternative to enhance performance while maintaining cost-effectiveness. With the advancement of fabrication and bonding technologies, wafer-level 3D integration enables fine-grain 3D interconnects that maximize the benefits in power, performance, and area (PPA). However, a multitude of challenges have obstructed traditional electronic design automation (EDA) methodologies for 3D IC implementations. This paper surveys the major challenges in the physical design of advanced 3D ICs. We provide a comprehensive review of existing solutions, analyzing their advantages and disadvantages in depth. Finally, we discuss open problems and research opportunities in the development of native 3D EDA tools.

CCS Concepts

• **Hardware** → **Physical synthesis**.

Keywords

3D ICs; Physical Design; Partitioning; Power Delivery; Placement; Clock Delivery; Routing

ACM Reference Format:

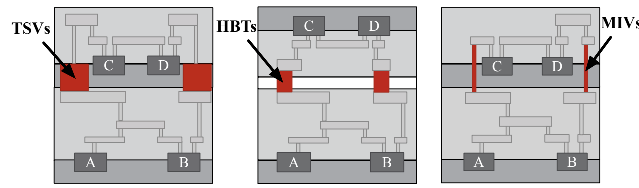
Yuxuan Zhao, Lancheng Zou, and Bei Yu. 2025. Invited: Physical Design for Advanced 3D ICs: Challenges and Solutions . In *Proceedings of the 2025 International Symposium on Physical Design (ISPD '25), March 16–19, 2025, Austin, TX, USA*. ACM, New York, NY, USA, 8 pages. https://doi.org/10.1145/3698364.3709127

1 Introduction

Since Gordon Moore's seminal prediction in 1965 [1], the feature size of transistors has steadily decreased. This technology scaling has enabled the integration of more transistors on a single die, thereby boosting performance and reducing costs. However, the economic benefits of continuing More-Moore scaling are diminishing. The industry is facing significant cost increases in moving to more advanced technology nodes (2 nm and beyond), primarily due to the high expenses of extreme ultraviolet (EUV) lithography, masks, electronic design automation (EDA), *etc*. In response, More-than-Moore integration through 3D stacking emerges as a promising solution to these challenges.

ISPD '25, Austin, TX, USA

ACM ISBN 979-8-4007-1293-7/25/03
https://doi.org/10.1145/3698364.3709127



(a) TSV based **(b) Hybrid bonding** **(c) Monolithic**

Figure 1: The typical types of 3D ICs. (a) TSVs occupy large silicon area, resulting in low integration density. (b) Hybrid bonding uses direct Cu-Cu bonds with fine pitches and necessitates die alignment after parallel fabrication of dies. (c) Monolithic 3D ICs offer the highest integration density, featuring nanometer-scale pitches for 3D vias, but require sequential die fabrication, which incurs higher costs.

3D ICs improve power, performance, and area (PPA) by vertically stacking multiple smaller dies, which achieves higher transistor density and replaces long 2D interconnects with shorter inter-die connections. There are primarily three types of 3D ICs: through-silicon-via (TSV) based, hybrid bonding, and monolithic. Figure 1 shows the 3D stacking for typical 3D ICs. TSVs [2, 3], with 10 µm-scale pitch size, occupy a significant amount of silicon area and introduce large parasitics, limiting TSV-based 3D ICs to a few inter-die connections. Despite these constraints, TSV-based 3D ICs are commercially the most viable, benefiting specific designs such as high bandwidth memory (HBM) [4, 5]. Hybrid bonding 3D ICs [6, 7] consist of two pre-fabricated dies, where the top die is flipped and bonded to the bottom die using hybrid bonding terminals (HBTs) with 1 µm-scale pitches on the back-end-of-lines (BEOLs). The ease of fabrication and small size of bonding terminals enable high integration density at reduced costs. Notable implementations, such as AMD's Zen 4 [8] and Meta's AR SoC [9], have achieved significant performance enhancements through hybrid bonding. Monolithic 3D ICs [10, 11], using nanoscale monolithic inter-tier vias (MIVs), enable the highest integration density. However, the sequential fabrication of multiple dies at low temperatures makes monolithic 3D (M3D) technology expensive. Although further from commercialization compared to other types of 3D ICs, M3D offers the most substantial potential to exploit the third dimension in IC design[12].

EDA tools play a crucial role in the design and implementation of modern ICs, where the physical design process converts flattened netlists into GDSII files containing all the physical information. However, commercially available tools like Innovus [13] and IC Compiler [14] lack native support for placement and routing (PnR) in 3D IC designs. To address this limitation, two methodological approaches have emerged. Pseudo-3D flows such as [15–18] leverage the capabilities of commercial 2D PnR tools. In a pseudo-3D flow, 3D designs are constructed using an intermediate 2D design and then partitioned into multiple dies and routed to produce the final layouts. Partitioning algorithms such as [19, 20] are employed to accommodate the 3D aspects of placement within these workflows. While pseudo-3D flows benefit from mature congestion and timing optimization features of commercial tools, they do not fully exploit the potential of fine-grain 3D interconnects found in wafer-level 3D integration, *i.e.*, hybrid bonding, and monolithic 3D ICs. On the other hand, true 3D implementations with specialized placers [21–24] optimize tier partitioning and cell locations within a 3D solution space, taking full advantage of the z-direction to minimize wirelength. However, these academic placers, still in the early stages of development, lack comprehensive support for final PPA optimization.

This paper surveys the major challenges and promising solutions in the physical design of advanced 3D ICs across five main stages: partitioning, power delivery, placement, clock delivery, and routing. We outline the key challenges associated with the physical implementation of 3D ICs at each design stage in Section 2. Following this, we provide a comprehensive review of existing methodologies, discussing their advantages and disadvantages in Section 3. Section 4 discusses research opportunities for developing native 3D EDA tools. Finally, Section 5 concludes the paper.

2 Challenges in Physical Design for Advanced 3D ICs

Over the past decades, both industry and academia have made significant efforts to develop physical design tools for 3D ICs, with partitioner, placer, and clock synthesis tools, *etc.*, being among the first to prototype 3D ICs. The following subsections will analyze the fundamental causes of challenges at each design stage and provide our insights for further investigation.

2.1 Partitioning

Tier partitioning is a crucial step in pseudo-3D flows [15–18] to map a 2D design into the 3D space. 3D partitioning algorithms distribute logic gates or blocks into different dies of a 3D IC to ensure that inter-die connections are either minimized or maintained below a specified upper limit. Most existing approaches employ partitioners originally designed for 2D ICs, aiming to minimize the number of cuts and balance the area of each partition.

However, these partitioning algorithms do not adequately address two major challenges. Firstly, minimizing the number of cuts does not necessarily optimize the number of inter-die connections [25, 26]. Current partitioners [27–29] neglect the distribution of partitions in the z-direction, treating cuts between any pair of partitions with equal importance. In reality, more 3D vias are required when two connected logic gates are located at non-adjacent dies. Therefore, 3D partitioners should penalize cuts between distant partitions to more effectively optimize the number of inter-die connections.

Moreover, current design flows [16–18] overlook critical technology and design-related parameters during partitioning, leading to PPA degradation. Traditional min-cut partitioners are not necessarily good for advanced 3D ICs as they cannot fully exploit high integration density in wafer-level 3D integration.

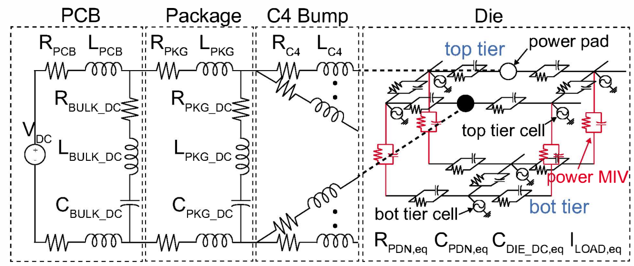


Figure 2: System-level PDN structure for M3D [35].

2.2 Power Delivery

Power delivery networks (PDNs) for 3D ICs are essential for ensuring efficient and reliable power distribution across the chip's stacked layers. Unlike traditional 2D ICs, 3D ICs consist of multiple layers stacked vertically, which introduces unique power distribution challenges. Power is supplied from the off-chip package through C4 bumps and distributed vertically via TSVs [30]. For power grid analysis, the on-chip circuit will be modeled as an RLC network, which, unlike the planar grids in 2D ICs, must account for vertical connections by the power TSVs [31].

Designing PDNs for 3D ICs presents complex challenges due to the intricate task of distributing power across multiple stacked layers and managing interactions between these layers. The integration of TSVs for power transmission introduces additional parasitics, such as resistance, capacitance, and inductance, that can degrade the chip's power integrity [32]. Moreover, the number of power TSVs is limited by the footprint of the 3D chip [32]. The architecture of 3D ICs also allows for power noise to transfer from one layer to another, necessitating effective decoupling strategies to handle the noise [33]. The high density of 3D ICs makes it challenging to find sufficient space for the necessary capacitors without affecting performance or area. To address these challenges, advanced modeling techniques and optimization algorithms are required to design robust PDNs that ensure stable power delivery and minimize noise. Additionally, thermal management becomes a critical concern, as power delivery interacts closely with heat dissipation in 3D ICs, necessitating co-optimization of power and thermal management [34].

In contrast to TSV-based 3D ICs, M3D ICs use power MIVs to connect the bottom metal layer of the top tier to the top metal layer of the bottom tier, creating a series resistive path across tiers [35]. The system-level PDN structure is illustrated in Figure 2. This results in a longer resistive path for the bottom-tier cells compared to TSV-based designs. Additionally, irregular power MIV placement, caused by cell blocking on the top tier, further complicates power delivery in M3D ICs. As a result, M3D ICs experience much higher static voltage drop, particularly in the bottom-tier cells.

2.3 Placement

While the adoption of netlist partitioning and die-by-die 2D placement significantly simplifies the complexity of the problem [16–18], the quality of placement is limited by the initial partitioning. Furthermore, die-by-die 2D placement does not take into account the vertical connections between dies. Consequently, cells that belong

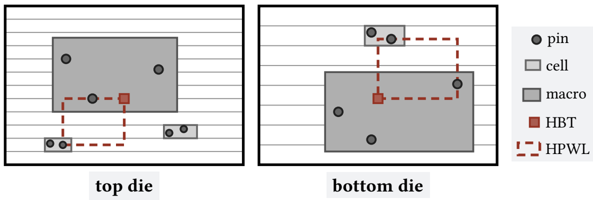


Figure 3: Illustration of die-to-die (D2D) HPWL [24]. D2D HPWL of a 3D net is the sum of the wirelength of the top net and bottom net. Hybrid bonding terminals (HBTs) are on the top-most layer for both dies.

to the same 3D net may end up being positioned far away from each other, resulting in increased wirelength.

True-3D placers [21–24] optimize cell partitioning and locations simultaneously. Compared to the pseudo-3D flows, these placers explore a substantially larger solution space to optimize wirelength. However, the implementation of 3D placers faces two significant challenges. Firstly, unlike 2D placement, which is solved within a continuous planar space, the z-direction in 3D placement is highly discrete, complicating the optimization problem. Secondly, the wirelength model for 3D ICs must be carefully designed to accurately reflect the wirelength. Traditional half-perimeter wirelength (HPWL) model is unsuitable as it does not account for the locations of 3D vias, which are crucial for evaluating total wirelength. Recent IC-CAD contests [36, 37] have addressed the die-to-die HPWL for hybrid bonding 3D ICs. Figure 3 illustrates how the location of hybrid bonding terminals is considered in the calculation of die-to-die HPWL.

2.4 Clock Delivery

Clock delivery network construction is a critical step in the physical design process that ensures the clock signal is efficiently distributed to all sequential elements, such as flip-flops, across the chip [38]. The primary goal of clock delivery network design is to minimize clock skew while ensuring power efficiency and signal integrity. Traditional 2D clock design typically employs H-tree, X-tree, or mesh-based structures to achieve balanced clock distribution. Compared to 2D ICs, clock design for 3D ICs poses new challenges due to the usage of clock TSVs and the allocation of sequential elements.

3D clock tree design must carefully account for the placement and utilization of clock TSVs to fully leverage the benefits of 3D integration. Underusing clock TSVs may fail to reduce wirelength effectively, limiting the potential advantages of shorter interconnects in 3D ICs. Conversely, overusing clock TSVs can introduce excessive parasitic capacitance, leading to increased power consumption, degraded signal integrity, and higher clock latency [39]. The clock delivery network is usually designed after the placement stage, thus the the allocation of the clock TSVs should be aware of the locations of the signal TSVs [40]. In addition, since P/G TSVs will also be used for the power delivery network, their placement also needs to be considered during the clock tree synthesis to prevent overlap.

The additional z-axis dimension in 3D ICs not only expands the design space for clock tree structures but also introduces new challenges due to increased complexity. The placement of sequential elements, the partitioning strategy, and the distribution of the clock tree across different tiers critically influence the TSV utilization, wirelength, and clock skew. An effective partitioning strategy is crucial for ensuring cross-tier synchronization, minimizing interconnect overhead, and accommodating design constraints such as power and thermal constraints [41].

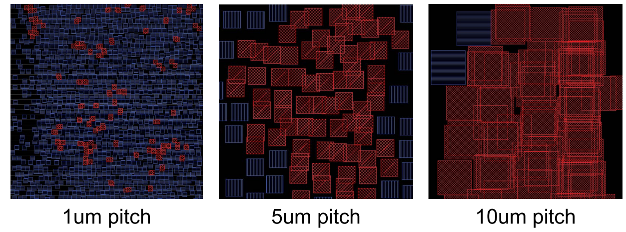


Figure 4: 3D via overlaps shown in red at various pitch values [44].

2.5 Routing

Routing involves connecting all the pins on a chip using wires across multiple metal layers. For 3D ICs, this process can be efficiently executed using conventional 2D routers, which are designed to accommodate the vertical stacking of metal layers. However, unlike the nanoscale vias between metal layers, inter-die connections are significantly larger and require specialized considerations during the routing phase.

In TSV-based 3D ICs, early work [42] uses a single TSV for a 3D net to reduce TSV number, but Kim *et al.* [43] demonstrate that employing multiple TSVs per net significantly improves the wirelength for 3D nets. Furthermore, in wafer-level bonded 3D ICs, which offer higher integration density, the effective utilization of 3D vias is crucial to achieving shorter wirelength. Despite these advances, current 3D PnR flows [15–18] fail to observe the 3D via spacing rules when realistic pitch values are used, as shown in Figure 4. To address this, a 3D via legalization stage [44] is necessary to reduce violations while maintaining minimal impact on wirelength.

3 Solutions for 3D IC Physical Design

In this section, we review the historical as well as the state-of-the-art solutions for challenges in each design stage. Additionally, we provide a detailed analysis of their advantages and disadvantages.

3.1 Partitioning

Most 3D partitioning algorithms [16–18] employ min-cut partitioners [28] to recursively partition a flattened netlist across an arbitrary number of 3D layers. However, these methods often overlook crucial design-related and physical information, leading to suboptimal partitioning outcomes. To address these deficiencies, recent advancements have introduced design-aware partitioners [15, 45] and machine learning-based approaches [20], which more effectively leverage the high integration density of advanced 3D ICs.

Min-cut partitioner. Circuit partitioning is a well-known NP-hard problem, and several effective heuristics were developed, such as Kernighan-Lin (KL) algorithm [27] and Fiduccia-Mattheyses (FM) algorithm [28]. These algorithms utilize selective cell swapping to iteratively reduce cut size. To address scalability issues in large

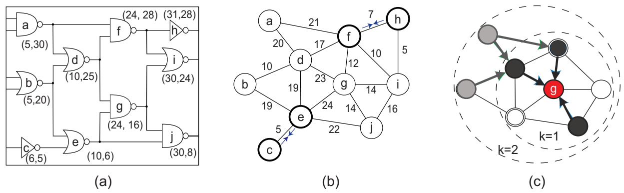


Figure 5: The framework of TP-GNN [20]. (a) Input netlist with 2D physical implementation. (b) Hierarchy-aware edge contractions on the transformed graph. (c) Sampling and aggregating features from its k-hop neighbors.

circuits, multi-level partitioners [29, 46, 47] decompose the partitioning task into hierarchical levels. Initially, the netlist is coarsened into larger blocks, which are then partitioned using the KL or FM algorithms, followed by a refinement phase to optimize the outcomes. However, traditional min-cut partitioners often overlook the distribution of vertical partitions, where cuts between non-adjacent dies necessitate increased use of 3D vias. Ababei *et al.*[25] address this issue by mapping the partition generated by hMetis[29] into a matrix and introducing a heuristic for matrix manipulation to assign layer indices. Sawicki *et al.* [26] enhance the hMetis solution through a simulated annealing algorithm, achieving a 15% reduction in 3D vias compared to the initial solution. Nevertheless, without incorporating design-related and physical information, min-cut partitioners may not fully utilize the benefits offered by advanced 3D ICs.

Design-aware partitioner. Designs like microprocessors and accelerators typically feature large memory blocks and clear hierarchical structures [48, 49]. Logic-on-memory partitioning [45, 50, 51] is a straightforward strategy to separate the logic and memory blocks into different dies. In modern microprocessors, caches often occupy more than 50% of the total footprint, leading to long logic-to-memory interconnects. By relocating low-level and large-size cache blocks to the memory die [51], these interconnects can be effectively implemented using 3D vias, and signal routing in logic modules is free from memory block obstructions. To better utilize the hierarchical information, Cascade-2D [15] performs partitioning at the register-transfer level, rather than at the gate-level netlist. With the understanding of micro-architecture organization, functional modules that have tight timing constraints, such as data path units and register banks, are placed on adjacent dies and connected with 3D vias to reduce wirelength. Design-aware partitioners, which typically produce a few inter-die connections, are most effective for designs that exhibit a clear hierarchy.

Machine learning-based partitioner. Graph neural networks (GNNs) have emerged as a powerful approach in modeling and predicting intricate outcomes in EDA problems [52–58]. TP-GNN [20] introduces an innovative unsupervised graph-learning-based approach for tier partitioning, effectively bridging the gap between initial partitioning and final PPA. The framework integrates multiple features including gate-level timing, topological information, and design hierarchy to address the PPA degradation issues in current 3D partitioners. Figure 5 illustrates the framework of TP-GNN. Starting with a 2D physical implementation of a design, the netlist hypergraph is converted into a clique-based graph through a hierarchy-aware edge contraction algorithm. Subsequently, the framework employs GNNs to perform graph representation learning, generating node embeddings that are utilized by a weighted k-means clustering algorithm to produce final partitions. However, layout modality with congestion information is neglected by this framework. Incorporating multimodal fusion techniques [54, 59] may further enhance the partitioning performance.

3.2 Power Delivery

In order to address the diverse challenges in 3D PDN construction, many researchers have been actively exploring innovative solutions.

P/G TSVs are critical components of the 3D PDN, and their placement and design require careful optimization to ensure a reliable power supply in 3D ICs. An MILP-based approach is proposed to simultaneously optimize the positions, the total number, and the sizing of P/G TSVs [31]. Additionally, exploring the configuration of C4 bumps is also crucial for enhancing 3D PDN performance. Researchers perform detailed architectural analysis [33] to study the IR drop and $L\frac{di}{dt}$ drop with various design choices, including TSV size, TSV and C4 granularity. A look-ahead mechanism [60–62] can be implemented to predict the impact of future stages of the design process. Moreover, strategies for floorplanning and P/G co-synthesis [63] have been developed, which creates an effective floorplan and P/G network that minimize area, wirelength, and IR drop in 3D ICs.

Accurate modeling methods are necessary to guide the design of a robust 3D PDN. For static IR drop analysis, most methods represent TSV parasitics using a single resistor. The resistance value is typically calculated under the assumption of a uniform current distribution across the TSV, which simplifies the complexities of the actual scenario. A graph attention network-based framework is adopted to achieve efficient analysis of current crowding and IR drop in F2F 3D IC TSVs [64]. Hetero-3D [65] implements Gaussian process regression (GPR) to predict the routing cost and IR drop of the PDN design. Moreover, a constrained optimization problem is formulated to minimize the total wire resistance while ensuring compliance with IR drop constraints.

Ensuring reliable power delivery in M3D ICs is a significant challenge, primarily due to their high power density and current demand per unit area. Moreover, the interconnects in M3D ICs are highly susceptible to electromigration (EM) and stress migration (SM), further complicating the design of an effective PDN. Genetic programming (GP) [66] is employed to explore the design space and optimize the PDN for reliable operation. The goal is to minimize voltage drop on power supply rails while adhering to constraints related to resilience against EM and SM.

3.3 Placement

The 3D placement problem involves assigning positions (x_i, y_i, z_i) to each cell to minimize total wirelength, while ensuring cells do not overlap and the number of 3D vias remains below a set threshold. To tackle the discrete layer assignment problem in z-direction, transformation-based placers [67, 68] use space transformation techniques to map a 2D placement to a 3D solution. Current pseudo-3D placers [15–18, 69] rely on tier partitioning and reuse 2D placement engine to simplify the problem. In contrast, 3D analytical

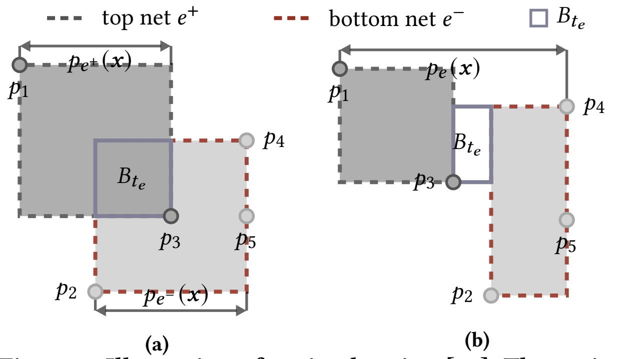


Figure 6: Illustration of optimal region [24]. The optimal region B_{t_e} is the region bounded by the median values of the top net box and bottom net box. Hybrid bonding terminal t_e placed outside B_{t_e} will introduce extra wirelength. (a) If top net and bottom net boxes overlap in x-axis, the minimal x-axis D2D HPWL is $p_{e^+}(x) + p_{e^-}(x)$. (b) If top net and bottom net boxes have no overlap in x-axis, the minimal x-axis D2D HPWL is $p_e(x)$.

placers [21–24, 70] relax the discrete layer assignment and adopt analytical approaches.

Transformation-based placer. T3Place [67] introduces effective transformation techniques such as local stacking, folding-2, folding-4, and window-based stacking/folding. After transformation, a relaxed conflict-net graph-based layer assignment is employed to refine the 3D placement. However, this transformed 3D placement solution relies solely on the initial 2D placement and does not optimize 3D wirelength. To address this limitation, HyPlace-3D [68] employs a 2D-to-3D analytical placement approach, distributing cells and 3D vias within an expanded view to optimize 3D wirelength. Additionally, pseudo nets are introduced for marginal cells to further minimize wirelength. Nonetheless, transformation-based placers, which convert the 3D space into expanded 2D layouts, inherently restrict the solution space, leading to suboptimal results.

Pseudo-3D placer. Pseudo-3D placers can be categorized into three distinct types: partitioning-first flow [15], partitioning-last flow [16–18], and alternating partitioning and placement strategies [69]. Cascade-2D [15] initially employs a design-aware partitioning technique to divide the design into two dies. Subsequently, it performs a trial placement for each die to determine the locations of 3D vias, which then serve as anchor cells during subsequent die-by-die placements, thereby reducing the wirelength of 3D nets. In contrast, partitioning-last flows [16–18] first perform 2D placement on a compressed 2D layout. Based on the physical information, a bin-based FM partitioning algorithm is utilized to distribute cells within a bin to different dies, so that the property of 2D placement solution is preserved. However, deciding the order of partitioning and placement is like a chicken-and-egg dilemma. To address this issue, iPL-3D [69] models the problem as a bilevel programming problem, alternately optimizing cell partitioning and placement. This method allows partitioning to provide an initial solution for placement, which in turn estimates the overall objective for subsequent partitioning. While this strategy enhances final performance, it offers only a local view of 3D placement during each phase.

Analytical placer. Analytical placers perform placement in a 3D cuboid region, utilizing smoothed wirelength and density models. NTU-3D [21] adopts the weighted-average wirelength model and bell-shaped density model to minimize total HPWL wirelength while observing maximum bin density constraints. White space for TSV placement is considered in density calculation during 3D global placement. ePlace-3D [70] introduces an innovative electrostatic force-based density model called eDensity-3D. The overall placement region is modeled as a 3D electrostatic field, with each cell represented as a positively charged cuboid. The electrostatic force will push cells away from high-density regions, thereby achieving a uniform density distribution. However, the traditional HPWL wirelength is inconsistent with the actual 3D wirelength, where the locations of 3D vias must be considered. Liao *et al.* [23] propose a bistratal wirelength model to accurately reflect the die-to-die HPWL in hybrid bonding 3D ICs. The authors prove that D2D HPWL is no less than traditional HPWL, and D2D HPWL is minimized when the 3D via is placed within its optimal region, which is defined by the median values of the top and bottom net boxes, as depicted in Figure 6. However, this model, specifically tailored for hybrid bonding 3D ICs with two dies, requires modifications to accommodate 3D ICs with additional tiers.

3.4 Clock Delivery

3D clock tree synthesis (CTS) typically involves a series of tasks: (1) abstract tree topology generation; (2) layer assignment of tree nodes; (3) clock tree routing (embedding); and (4) clock buffer insertion [41, 71]. The abstract tree denotes the hierarchical connectivity between a centralized clock source and all sequential elements. Layer assignment determines the vertical positions of the tree nodes. Clock tree routing determines the planar locations of the intermediate clock tree nodes and clock TSVs, which is carried out alongside clock buffer insertion to satisfy the skew constraint.

For abstract tree generation, 3D-MMM is developed to recursively divide the given sink set into two subsets until each sink belongs to its own set [39], which is an extension of the method of means and medians (MMM) algorithm [38]. In addition, its partitioning direction (*i.e.*, X/Y or Z direction) is determined by the available TSV counts. Different from 3D-MMM [39], the partitioning direction is chosen based on the HPWL of subsets in [71]. A TSV-aware topology generation algorithm [72], which uses a sorting method to cluster clock sinks, is introduced to consider the density distribution of allocated TSVs, the parasitic, and coupling effects. TSV equivalent wire length (TEWL) is employed to evaluate the overhead of TSVs. This approach can achieve a balanced distribution of TSVs, adheres to manufacturing constraints, and improves reliability.

In the process of clock tree routing, the objective is to minimize the total wirelength of the clock tree and clock skew. 3D deferred layer embedding (DLE) and 3D deferred merge embedding (DME) are proposed to minimize the TSV cost for a tree topology and total wirelength with the consideration of the delay effect of TSVs [74]. Zhao *et al.* [40] present a TSV-induced obstacle-aware CTS method for 3D ICs, addressing challenges posed by placement and routing obstacles created by various TSVs (signal, power/ground, and clock). The proposed algorithm avoids TSV-induced overlaps while

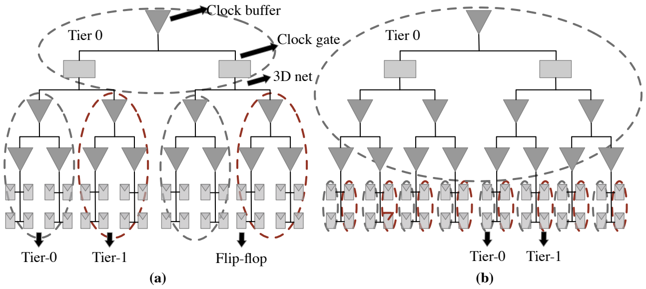


Figure 7: Two different types of 3D CTS (a) One clock tree per tier for each gating group (source-level), and (b) The entire backbone is fixed onto tier 0 (leaf-level) [73].

maintaining minimal wirelength and power overhead, achieving a reliable buffered clock tree. A TSV array is more practical and manufacturable than layouts with irregularly positioned TSVs. However, the utilization of TSVs in a 3D clock network has a significant impact on the final clock power due to the limited availability of TSV resources. To address this, Zhao *et al.* [75] propose a decision-tree-based clock synthesis (DTCS) method that efficiently explores the entire solution space to optimize TSV array utilization, enabling the generation of low-power and reliable clock networks.

Commercial tools currently lack support for 3D CTS. To address this, Shrunk-2D [73] implements customized scripts that function as plugins to enable 3D CTS in these tools. Traditional 3D CTS methods involve constructing a separate clock tree for each die and connecting them with a single MIV. In contrast, a new approach introduced by Shrunk-2D places the main clock backbone on the bottom tier, allowing for flexible placement of the leaf-level flip-flops across other tiers to ensure balanced area distribution. The two different types of 3D CTS are illustrated in Figure 7. Additionally, Vanna-Iampikul *et al.* [76] introduce a GNN-based multi-bit flip-flops clustering (MBFC) technique to optimize the clock network in terms of power and performance.

3.5 Routing

Existing commercial routers can be used to perform routing for 3D ICs by stacking metal layers of multiple dies. However, the runtime increases significantly as the complexity of the design goes up. To address this issue, several studies [43, 77–79] suggest dividing the 3D routing process into two distinct sub-steps: 3D via insertion and die-by-die routing.

Kim *et al.* [43] propose a 3D rectilinear steiner minimum tree (RSMT) construction algorithm to find optimal routing topologies for 3D nets. Based on the 2D RSMT for the 3D nets, the authors use a bottom-up breath-first algorithm to calculate the die span for each steiner point. TSVs are inserted according to the die span of adjacent steiner points. Inspired by the lookup table-based RSMT algorithm FLUTE [80], Dewan *et al.* [79] construct all multilayer monolithic RSMTs on the 3D Hanan grid. The 3D RSMT has the shortest planar wirelength with a minimum number of 3D vias, which can be used to perform 3D via insertion.

Current 3D PnR flows [15–18] often fail to adhere to 3D via spacing rules when realistic pitch values are applied. A 3D via legalization stage [44, 81] is needed to reduce the violations. Pentapati *et al.* [44] propose a force-based solver and a bipartite-matching algorithm with Bayesian optimization to find legal positions for 3D vias post-routing. Conversely, BTAssign [81] utilizes an iterative hierarchical bipartite-matching algorithm to assign 3D vias to legal positions before routing. Although these approaches solve the legality issue, they separate the 3D via legalization and routing tree construction, resulting in suboptimal results.

4 Open Problems and Opportunities for Research

Although the literature has extensively explored various EDA challenges, current solutions often involve adapting existing commercial 2D tools with specialized scripts to facilitate 3D analysis. To fully exploit the PPA benefits of 3D stacked technology, large-scale native 3D EDA tools must be implemented. Below, we provide some promising research directions in each physical design stage:

- 3D partitioners need to integrate both design-related and physical information to achieve effective partitioning outcomes. Machine learning approaches, particularly multimodal fusion techniques that combine graph and layout information, show great promise in enabling PPA-aware partitioning at the early stage.
- PDN design should be considered with other design stages, especially P/G TSVs need to be co-optimized with other TSVs (signal, clock) to balance the utilization.
- In addition to minimizing wirelength, a 3D placer must also account for other placement constraints, including blockages, routability, timing, manufacturability, and reliability.
- Managing clock skew effectively across multiple layers remains a critical challenge in 3D clock tree synthesis, especially as the complexity increases with the number of stacked dies.
- The separation of intra-die and inter-die routing substantially compromises the quality of routing solutions. A native 3D router, designed to simultaneously determine legal 3D via locations and routing tree structures, holds promise for delivering superior outcomes.

The whole physical design flow for 3D ICs consists of complex procedures and adjustable parameters. Design engineers must employ physical design tools iteratively to meet the PPA targets. Design space exploration methods, including Bayesian optimization [82–86] and reinforcement learning [87], can significantly accelerate this process. Additionally, large language models (LLMs) have the potential to automatically generate design flow scripts [88, 89], further streamlining the control of the physical design process for 3D ICs.

5 Conclusion

3D ICs have shown promising enhancements in power, performance, and area (PPA), independent of the costly scaling of transistors. However, the increased complexity of design space due to 3D integration poses significant challenges to the physical design domain, including partitioning, power delivery, placement, clock delivery, and routing. This paper analyzes the root causes of these challenges in 3D IC physical design and reviews state-of-the-art solutions at each design stage. We conclude that design automation for 3D ICs remains a formidable challenge and has not yet been fully addressed. The development of large-scale native 3D EDA tools, coupled with advanced design space exploration agents, is crucial for fully realizing the potential of this promising technology frontier.

Acknowledgment

This work is partially supported by The Research Grants Council of Hong Kong SAR (No. CUHK14210723 and No. CUHK14211824), and the MIND project (MINDXZ202404).

References

[1] G. E. Moore, "Cramming more components onto integrated circuits," *Proceedings of the IEEE*, vol. 86, no. 1, pp. 82–85, 1998.

[2] M. Motoyoshi, "Through-silicon via (TSV)," *Proceedings of the IEEE*, vol. 97, no. 1, pp. 43–48, 2009.

[3] T. Lu, C. Serafy, Z. Yang, S. K. Samal, S. K. Lim, and A. Srivastava, "TSV-based 3-D ics: Design methods and tools," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 36, no. 10, pp. 1593–1619, 2017.

[4] D. U. Lee, K. W. Kim, K. W. Kim, H. Kim, J. Y. Kim, Y. J. Park, J. H. Kim, D. S. Kim, H. B. Park, J. W. Shin *et al.*, "25.2 A 1.2 V 8Gb 8-channel 128GB/s high-bandwidth memory (HBM) stacked DRAM with effective microbump I/O test methods using 29nm process and TSV," in *IEEE International Solid-State Circuits Conference (ISSCC).* IEEE, 2014, pp. 432–433.

[5] J.-Y. Kim, T. Kim, J. You, K. Kim, B. M. Moon, K. Sohn, and S.-O. Jung, "An energy-efficient design of TSV I/O for HBM with a data rate up to 10 Gb/s," *IEEE Journal Solid-State Circuits*, vol. 58, no. 11, pp. 3242–3252, 2023.

[6] C. Premachandran, J. Lau, L. Xie, A. Khairyanto, K. Chen, M. E. P. Pa, M. Chew, and W. K. Choi, "A novel, wafer-level stacking method for low-chip yield and non-uniform, chip-size wafers for MEMS and 3D SIP applications," in *IEEE Electronic Components and Technology Conference.* IEEE, 2008, pp. 314–318.

[7] Y. Chen, C. Yang, C. Kuo, M. Chen, C. Tung, W. Chiou, and D. Yu, "Ultra high density soic with sub-micron bond pitch," in *IEEE Electronic Components and Technology Conference.* IEEE, 2020, pp. 576–581.

[8] B. Munger, K. Wilcox, J. Sniderman, C. Tung, B. Johnson, R. Schreiber, C. Henrion, K. Gillespie, T. Burd, H. Fair *et al.*, ""Zen 4": The AMD 5nm 5.7 GHz x86-64 microprocessor core," in *IEEE International Solid-State Circuits Conference (ISSCC)*, 2023, pp. 38–39.

[9] T. F. Wu, H. Liu, H. E. Sumbul, L. Yang, D. Baheti, J. Coriell, W. Koven, A. Krishnan, M. Mittal, M. T. Moreira *et al.*, "11.2 A 3D integrated prototype system-on-chip for augmented reality applications using face-to-face wafer bonded 7nm logic at $< 2\mu$m pitch with up to 40% energy reduction at iso-area footprint," in *IEEE International Solid-State Circuits Conference (ISSCC)*, vol. 67. IEEE, 2024, pp. 210–212.

[10] P. Batude, T. Ernst, J. Arcamone, G. Arndt, P. Coudrain, and P.-E. Gaillardon, "3-D sequential integration: A key enabling technology for heterogeneous co-integration of new function with CMOS," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 2, no. 4, pp. 714–722, 2012.

[11] S. S. K. Pentapati and S. K. Lim, "Heterogeneous monolithic 3D ICs: Eda solutions, and power, performance, cost tradeoffs," in *ACM/IEEE Design Automation Conference (DAC).* IEEE, 2021, pp. 925–930.

[12] L. Brunet, S. Reboh, T. Januel, X. Garros, T. M. Frutuoso, M. Cassé, B. Sklénard, L. Brévard, M. Ribotta, A. Magalhaes-Lucas *et al.*, "Record performance of 500° c low-temperature nMOSFETs for 3D sequential integration using a smart cut tm layer transfer module," in *2021 Symposium on VLSI Technology.* IEEE, 2021, pp. 1–2.

[13] Cadence, "Cadence Innovus implementation system," https://www.cadence.com, 2024, accessed: 2024-12-04.

[14] Synopsys, "Synopsys IC Compiler," https://www.synopsys.com, 2024, accessed: 2024-12-04.

[15] K. Chang, S. Sinha, B. Cline, R. Southerland, M. Doherty, G. Yeric, and S. K. Lim, "Cascade2D: A design-aware partitioning approach to monolithic 3D IC with 2D commercial tools," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2016, pp. 1–8.

[16] S. Panth, K. Samadi, Y. Du, and S. K. Lim, "Shrunk-2-D: A physical design methodology to build commercial-quality monolithic 3-D ICs," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 36, no. 10, pp. 1716–1724, 2017.

[17] B. W. Ku, K. Chang, and S. K. Lim, "Compact-2D: A physical design methodology to build commercial-quality face-to-face-bonded 3D ICs," in *ACM International Symposium on Physical Design (ISPD)*, 2018, pp. 90–97.

[18] S. S. K. Pentapati, K. Chang, V. Gerousis, R. Sengupta, and S. K. Lim, "Pin-3D: A physical synthesis and post-layout optimization flow for heterogeneous monolithic 3D ICs," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2020, pp. 1–9.

[19] S. Panth, K. Samadi, Y. Du, and S. K. Lim, "Placement-driven partitioning for congestion mitigation in monolithic 3d ic designs," in *ACM International Symposium on Physical Design (ISPD)*, 2014, pp. 47–54.

[20] Y.-C. Lu, S. S. K. Pentapati, L. Zhu, K. Samadi, and S. K. Lim, "Tp-gnn: A graph neural network framework for tier partitioning in monolithic 3d ics," in *ACM/IEEE Design Automation Conference (DAC).* IEEE, 2020, pp. 1–6.

[21] M.-K. Hsu, V. Balabanov, and Y.-W. Chang, "TSV-aware analytical placement for 3-D IC designs based on a novel weighted-average wirelength model," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 32, no. 4, pp. 497–509, 2013.

[22] Y.-J. Chen, C.-H. Hsieh, P.-H. Su, S.-H. Chen, and Y.-W. Chang, "Mixed-size 3d analytical placement with heterogeneous technology nodes," in *ACM/IEEE Design Automation Conference (DAC)*, 2024, pp. 1–6.

[23] P. Liao, Y. Zhao, D. Guo, Y. Lin, and B. Yu, "Analytical die-to-die 3d placement with bistratal wirelength model and gpu acceleration," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2023.

[24] Y. Zhao, P. Liao, S. Liu, J. Jiang, Y. Lin, and B. Yu, "Analytical heterogeneous die-to-die 3d placement with macros," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[25] C. Ababei, Y. Feng, B. Goplen, H. Mogal, T. Zhang, K. Bazargan, and S. Sapatnekar, "Placement and routing in 3d integrated circuits," *IEEE Design & Test of Computers*, vol. 22, no. 6, pp. 520–531, 2005.

[26] S. Sawicki, G. Wilke, M. Johann, and R. Reis, "A cells and i/o pins partitioning refinement algorithm for 3d vlsi circuits," in *IEEE International Conference on Electronics, Circuits and Systems.* IEEE, 2009, pp. 852–855.

[27] B. W. Kernighan and S. Lin, "An efficient heuristic procedure for partitioning graphs," *The Bell system technical journal*, vol. 49, no. 2, pp. 291–307, 1970.

[28] C. M. Fiduccia and R. M. Mattheyses, "A linear-time heuristic for improving network partitions," in *ACM/IEEE Design Automation Conference (DAC)*, 1988, pp. 241–247.

[29] G. Karypis, R. Aggarwal, V. Kumar, and S. Shekhar, "Multilevel hypergraph partitioning: Application in vlsi domain," in *ACM/IEEE Design Automation Conference (DAC)*, 1997, pp. 526–529.

[30] N. H. Khan, S. M. Alam, and S. Hassoun, "System-level comparison of power delivery design for 2d and 3d ics," in *IEEE International 3D Systems Integration Conference (3DIC)*, 2009, pp. 1–7.

[31] S. Wang, F. Firouzi, F. Oboril, and M. B. Tahoori, "P/g tsv planning for ir-drop reduction in 3d-ics," in *IEEE/ACM Proceedings Design, Automation and Test in Eurpoe (DATE)*, 2014, pp. 1–6.

[32] P. Jain, T.-H. Kim, J. Keane, and C. H. Kim, "A multi-story power delivery technique for 3d integrated circuits," in *IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, 2008, pp. 57–62.

[33] N. H. Khan, S. M. Alam, and S. Hassoun, "Power delivery design for 3-d ics using different through-silicon via (tsv) technologies," *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, vol. 19, no. 4, pp. 647–658, 2010.

[34] S. Panth, K. Samadi, Y. Du, and S. K. Lim, "Tier-partitioning for power delivery vs cooling tradeoff in 3d vlsi for mobile applications," in *ACM/IEEE Design Automation Conference (DAC)*, 2015, pp. 1–6.

[35] K. Chang, S. Das, S. Sinha, B. Cline, G. Yeric, and S. K. Lim, "Frequency and time domain analysis of power delivery network for monolithic 3d ics," in *IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, 2017, pp. 1–6.

[36] K.-S. Hu, I.-J. Lin, Y.-H. Huang, H.-Y. Chi, Y.-H. Wu, and C.-F. C. Shen, "2022 ICCAD CAD Contest Problem B: 3D placement with D2D vertical connections," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2022, pp. 1–5.

[37] K.-S. Hu, H.-Y. Chi, I.-J. Lin, Y.-H. Wu, W.-H. Chen, and Y.-T. Hsieh, "2023 ICCAD CAD Contest Problem B: 3D placement with macros," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023.

[38] M. A. Jackson, A. Srinivasan, and E. S. Kuh, "Clock routing for high-performance ics," in *ACM/IEEE Design Automation Conference (DAC)*, 1991, pp. 573–579.

[39] X. Zhao, J. Minz, and S. K. Lim, "Low-power and reliable clock network design for through-silicon via (tsv) based 3d ics," *IEEE Transactions on Components, Packaging and Manufacturing Technology (TCPMT)*, vol. 1, no. 2, pp. 247–259, 2010.

[40] X. Zhao and S. K. Lim, "Through-silicon-via-induced obstacle-aware clock tree synthesis for 3d ics," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC).* IEEE, 2012, pp. 347–352.

[41] J. Minz, X. Zhao, and S. K. Lim, "Buffered clock tree synthesis for 3d ics under thermal variations," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*, 2008, pp. 504–509.

[42] M.-C. Tsai, T.-C. Wang, and T. Hwang, "Through-silicon via planning in 3-d floorplanning," *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, vol. 19, no. 8, pp. 1448–1457, 2010.

[43] D. H. Kim, R. O. Topaloglu, and S. K. Lim, "Block-level 3d ic design with through-silicon-via planning," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC).* IEEE, 2012, pp. 335–340.

[44] S. Pentapati, A. Agnesina, M. Brunion, Y.-H. Huang, and S. K. Lim, "On legalization of die bonding bumps and pads for 3d ics," in *ACM International Symposium on Physical Design (ISPD)*, 2023, pp. 62–70.

[45] L. Bamberg, A. García-Ortiz, L. Zhu, S. Pentapati, S. K. Lim *et al.*, "Macro-3D: A physical design methodology for face-to-face-stacked heterogeneous 3D ICs," in *IEEE/ACM Proceedings Design, Automation and Test in Eurpoe (DATE)*, 2020, pp. 37–42.

[46] A. E. Caldwell, A. B. Kahng, and I. L. Markov, "Improved algorithms for hypergraph bipartitioning," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*, 2000, pp. 661–666.

[47] S. Schlag, T. Heuer, L. Gottesbüren, Y. Akhremtsev, C. Schulz, and P. Sanders, "High-quality hypergraph partitioning," *ACM Journal of Experimental Algorithmics*, vol. 27, pp. 1–39, 2023.

[48] A. B. Kahng, R. Varadarajan, and Z. Wang, "Hier-rtlmp: A hierarchical automatic macro placer for large-scale complex ip blocks," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2023.

[49] A. B. Kahng and Z. Wang, "Dg-replace: A dataflow-driven gpu-accelerated analytical global placement framework for machine learning accelerators," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[50] S. Pentapati, L. Zhu, L. Bamberg, A. García-Ortiz, S. K. Lim *et al.*, "A logic-on-memory processor-system design with monolithic 3-d technology," *IEEE Micro*, vol. 39, no. 6, pp. 38–45, 2019.

[51] L. Zhu, L. Bamberg, A. Agnesina, F. Catthoor, D. Milojevic, M. Komalan, J. Ryckaert, A. Garcia-Ortiz, and S. K. Lim, "Heterogeneous 3d integration for a risc-v system with stt-mram," *IEEE Computer Architecture Letters*, vol. 19, no. 1, pp. 51–54, 2020.

[52] Z. He, Z. Wang, C. Bai, H. Yang, and B. Yu, "Graph learning-based arithmetic block identification," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 2021, pp. 1–8.

[53] Z. Wang, C. Bai, Z. He, G. Zhang, Q. Xu, T.-Y. Ho, B. Yu, and Y. Huang, "Functionality matters in netlist representation learning," in *Proceedings of the 59th ACM/IEEE Design Automation Conference*, 2022, pp. 61–66.

[54] Z. Wang, S. Liu, Y. Pu, S. Chen, T.-Y. Ho, and B. Yu, "Restructure-tolerant timing prediction via multimodal fusion," in *ACM/IEEE Design Automation Conference (DAC)*. IEEE, 2023, pp. 1–6.

[55] X. Zhang, B. Zhu, F. Liu, Z. Wang, P. Xu, H. Xu, and B. Yu, "Disentangle, align and generalize: Learning a timing predictor from different technology nodes," in *ACM/IEEE Design Automation Conference (DAC)*, 2024, pp. 1–6.

[56] Z. Wang, W. Zhao, Y. Pu, L. Chen, W. W. K. Thong, W. Sheng, T.-Y. Ho, and B. Yu, "Parsgcn: Bridging the gap between emulation partitioning and scheduling," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[57] Z. Wang, C. Bai, Z. He, G. Zhang, Q. Xu, T.-Y. Ho, Y. Huang, and B. Yu, "Fgnn2: A powerful pre-training framework for learning the logic functionality of circuits," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[58] Z. Wang, Z. He, C. Bai, H. Yang, and B. Yu, "Efficient arithmetic block identification with graph learning and network-flow," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 42, no. 8, pp. 2591–2603, 2022.

[59] Y. Ye, P. Xu, L. Ren, T. Chen, H. Yan, B. Yu, and L. Shi, "Learning-driven physically-aware large-scale circuit gate sizing," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[60] S. Liu, Q. Sun, P. Liao, Y. Lin, and B. Yu, "Global placement with deep learning-enabled explicit routability optimization," in *IEEE/ACM Proceedings Design, Automation and Test in Eurpoe (DATE)*, 2021, pp. 1821–1824.

[61] S. Zheng, L. Zou, S. Liu, Y. Lin, B. Yu, and M. Wong, "Mitigating distribution shift for congestion optimization in global placement," in *ACM/IEEE Design Automation Conference (DAC)*, 2023, pp. 1–6.

[62] S. Zheng, L. Zou, P. Xu, S. Liu, B. Yu, and M. Wong, "Lay-net: Grafting netlist knowledge on layout-based congestion prediction," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 2023, pp. 1–9.

[63] P. Falkenstern, Y. Xie, Y.-W. Chang, and Y. Wang, "Three-dimensional integrated circuits (3d ic) floorplan and power/ground network co-synthesis," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*, 2010, pp. 169–174.

[64] Z. Yang, Z. Zhuang, B. Yu, T.-Y. Ho, M. D. F. Wong, and S. K. Lim, "Ml-based fine-grained modeling of dc current crowding in power delivery tsvs for face-to-face 3d ics," in *ACM International Symposium on Physical Design (ISPD)*, 2025.

[65] L. Zhu, J. Hu, G. Murali, and S. K. Lim, "Hetero-3d: Maximizing performance and power delivery benefits of heterogeneous 3d ics," in *IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, 2024, pp. 1–6.

[66] A. Koneru, A. Todri-Sanial, and K. Chakrabarty, "Reliable power delivery and analysis of power-supply noise during testing in monolithic 3d ics," in *IEEE VLSI Test Symposium (VTS)*, 2019, pp. 1–6.

[67] J. Cong, G. Luo, J. Wei, and Y. Zhang, "Thermal-aware 3d ic placement via transformation," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*. IEEE, 2007, pp. 780–785.

[68] J.-M. Lin, Y.-C. Lin, H. Kung, and W.-Y. Lin, "Hyplace-3d: A hybrid placement approach for 3d ics using space transformation technique," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 2023, pp. 1–8.

[69] X. Zhao, S. Chen, Y. Qiu, J. Li, Z. Huang, B. Xie, X. Li, and Y. Bao, "iPL-3D: A novel bilevel programming model for die-to-die placement," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023.

[70] J. Lu, H. Zhuang, I. Kang, P. Chen, and C.-K. Cheng, "ePlace-3D: Electrostatics based placement for 3D-ICs," in *ACM International Symposium on Physical Design (ISPD)*, 2016, pp. 11–18.

[71] T.-Y. Kim and T. Kim, "Clock tree embedding for 3d ics," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*, 2010, pp. 486–491.

[72] W. Liu, H. Du, Y. Wang, Y. Ma, Y. Xie, J. Quan, and H. Yang, "Tsv-aware topology generation for 3d clock tree synthesis," in *IEEE International Symposium on Quality Electronic Design (ISQED)*, 2013, pp. 300–307.

[73] S. A. Panth, K. Samadi, Y. Du, and S. K. Lim, "Design and cad methodologies for low power gate-level monolithic 3d ics," in *IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, 2014, pp. 171–176.

[74] T.-Y. Kim and T. Kim, "Clock tree synthesis for tsv-based 3d ic designs," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, vol. 16, no. 4, pp. 1–21, 2011.

[75] X. Zhao and S. K. Lim, "Tsv array utilization in low-power 3d clock network design," in *Proceedings of the 2012 ACM/IEEE international symposium on Low power electronics and design*, 2012, pp. 21–26.

[76] P. Vanna-Iampikul, Y.-C. Lu, D. E. Shim, and S. K. Lim, "Gnn-based multi-bit flip-flop clustering and post-clustering design optimization for energy-efficient 3d ics," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, vol. 28, no. 5, pp. 1–26, 2023.

[77] M. Pathak and S. K. Lim, "Performance and thermal-aware steiner routing for 3-d stacked ics," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 28, no. 9, pp. 1373–1386, 2009.

[78] P.-Y. Hsu, H.-T. Chen, and T. Hwang, "Stacking signal tsv for thermal dissipation in global routing for 3-d ic," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 33, no. 7, pp. 1031–1042, 2014.

[79] M. I. Dewan, S.-E. D. Lin, and D. H. Kim, "Construction of all multilayer monolithic rsmts and its application to monolithic 3d ic routing," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, vol. 29, no. 1, pp. 1–28, 2023.

[80] C. Chu and Y.-C. Wong, "FLUTE: Fast lookup table based rectilinear steiner minimal tree algorithm for VLSI design," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 27, no. 1, pp. 70–83, 2008.

[81] S. Liu, J. Jiang, Z. He, Z. Wang, Y. Lin, B. Yu, and M. Wong, "Routing-aware legal hybrid bonding terminal assignment for 3D face-to-face stacked ics," in *ACM International Symposium on Physical Design (ISPD)*, 2024, pp. 75–82.

[82] H. Geng, Q. Xu, T.-Y. Ho, and B. Yu, "Ppatuner: Pareto-driven tool parameter auto-tuning in physical design via gaussian process transfer learning," in *ACM/IEEE Design Automation Conference (DAC)*, 2022, pp. 1237–1242.

[83] H. Geng, T. Chen, Q. Sun, and B. Yu, "Techniques for cad tool parameter auto-tuning in physical synthesis: A survey," in *IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC)*. IEEE, 2022, pp. 635–640.

[84] S. Zheng, H. Geng, C. Bai, B. Yu, and M. D. Wong, "Boosting vlsi design flow parameter tuning with random embedding and multi-objective trust-region bayesian optimization," *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, vol. 28, no. 5, pp. 1–23, 2023.

[85] H. Geng, T. Chen, Y. Ma, B. Zhu, and B. Yu, "Ptpt: Physical design tool parameter tuning via multi-objective bayesian optimization," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, vol. 42, no. 1, pp. 178–189, 2022.

[86] P. Xu, S. Zheng, Y. Ye, C. Bai, S. Xu, H. Geng, T.-Y. Ho, and B. Yu, "Ranktuner: When design tool parameter tuning meets preference bayesian optimization," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 2024, pp. 1–7.

[87] H.-H. Hsiao, P. Vanna-Iampikul, Y.-C. Lu, and S. K. Lim, "Ml-based physical design parameter optimization for 3d ics: From parameter selection to optimization," in *ACM International Symposium on Physical Design (ISPD)*, 2024, pp. 1–6.

[88] H. Wu, Z. He, X. Zhang, X. Yao, S. Zheng, H. Zheng, and B. Yu, "Chateda: A large language model powered autonomous agent for eda," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.

[89] Y. Pu, Z. He, T. Qiu, H. Wu, and B. Yu, "Customized retrieval augmented generation and benchmarking for eda tool documentation qa," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*. IEEE, 2024, pp. 1–9.